

IGNOU BCS-011: Computer Basics and PC Software

Term-End Examination, December 2022 - Full Solved Paper

Time: 3 Hours Maximum Marks: 100

This solution booklet rewrites every question from the December 2022 BCS-011 paper and gives concise, exam-ready answers. Diagrams and flowcharts are redrawn using TikZ where useful.

Contents

Question 1

1(a) Do the following conversions.

Answer.

(i) **Convert $(37.5)_{10}$ to binary.**

Integer part: $37 = 32 + 4 + 1$, hence $37_{10} = 100101_2$.

Fractional part:

$$0.5 \times 2 = 1.0 \quad \Rightarrow \quad 0.5_{10} = 0.1_2.$$

Therefore,

$$\boxed{(37.5)_{10} = (100101.1)_2}.$$

(ii) **Convert $(100100101101)_2$ to hexadecimal.**

Group the binary number into four bits from the right:

$$1001 \ 0010 \ 1101.$$

Now $1001 = 9$, $0010 = 2$, and $1101 = D$. Therefore,

$$\boxed{(100100101101)_2 = (92D)_{16}}.$$

(iii) **Convert $(ABE)_{16}$ to binary.**

Each hexadecimal digit is replaced by four binary bits:

$$A = 1010, \quad B = 1011, \quad E = 1110.$$

Hence,

$$\boxed{(ABE)_{16} = (1010 \ 1011 \ 1110)_2}.$$

(iv) **Convert $(574)_8$ to decimal.**

$$(574)_8 = 5 \times 8^2 + 7 \times 8^1 + 4 \times 8^0 = 320 + 56 + 4 = 380.$$

Therefore,

$$\boxed{(574)_8 = (380)_{10}}.$$

1(b) What is main memory of the computer and why is it needed? Explain briefly the various types of main memory.

Answer. Main memory is the internal memory directly accessible by the CPU. It stores the currently running programs, data being processed, intermediate results and operating system routines required for execution. It is needed because secondary storage devices such as hard disks are much slower than the CPU. Main memory provides a fast working area between CPU and storage.

Types of main memory:

- (i) **RAM (Random Access Memory):** Read-write, volatile memory used to hold currently active programs and data. Its contents are lost when power is switched off.
- (ii) **ROM (Read Only Memory):** Non-volatile memory used to store permanent instructions such as boot firmware. Its contents remain even after power is off.
- (iii) **Cache memory:** Very high-speed memory placed between CPU and RAM. It stores frequently used instructions and data to reduce CPU waiting time.

1(c) Why are arrays used in programming? Explain with the help of an example of one-dimensional array.

Answer. An array is used to store multiple values of the same data type under one common name. Instead of declaring many separate variables, an array allows indexed access to a collection of related values. It is useful for marks, salaries, temperatures, roll numbers and similar lists.

Example: Suppose marks of five students are to be stored. In C language:

```
int marks[5] = {72, 65, 80, 58, 90};
```

Here, `marks` is a one-dimensional integer array. The first element is `marks[0]`, the second is `marks[1]`, and so on. To print all marks, a loop can be used:

```
for(i = 0; i < 5; i++)  
    printf("%d", marks[i]);
```

1(d) Differentiate between the following.

Answer.

(i) **Subroutine and Function**

Basis	Subroutine	Function
Meaning	A named block of code performing a task.	A named block of code that usually computes and returns a value.
Return value	May not return a value.	Normally returns a value.
Use	Used for action such as printing or updating.	Used for calculation such as finding sum or maximum.
Example	<code>displayResult()</code>	<code>int sum(int a,int b)</code>

(ii) **Modem and Router**

Basis	Modem	Router
Full form/role	Modulator-demodulator.	Device that forwards packets between networks.
Main job	Converts digital computer signals into suitable transmission signals and back.	Selects route and connects one network to another.
Use	Connects a computer/network to an ISP line.	Connects LAN to WAN or multiple networks.
Layer idea	Works mainly as access/communication equipment.	Works at network layer using IP addresses.

(iii) **URL and IP Address**

Basis	URL	IP Address
Meaning	Uniform Resource Locator.	Numeric logical address of a device.
Purpose	Specifies the address of a web resource.	Identifies a host/device on a network.
Readability	Human readable, <code>https://www.ignou.ac.in.</code>	Numeric, e.g. <code>192.168.1.10</code> .
Contains	Protocol, domain name, path.	Network part and host/device part.

1(e) What are communication ports and why are they needed? Explain parallel ports and serial ports.

Answer. Communication ports are physical or logical interfaces through which a computer communicates with external devices such as printers, scanners, keyboards, modems and storage devices. They are needed to connect peripherals and transfer data between the computer and external equipment.

Parallel port: A parallel port transmits several bits at the same time through multiple lines. It was commonly used for printers. It can transfer data faster over short distances but requires more wires and is less suitable for long-distance communication.

Serial port: A serial port transmits data one bit at a time over a single communication line. It was used for modems, mice and other serial devices. It is simpler, cheaper and suitable for longer distances, although traditional serial ports were slower than parallel ports. Modern USB is also serial in nature but much faster.

1(f) Why do computer systems need antivirus software? Which techniques do antivirus programs use to identify viruses?

Answer. Computer systems need antivirus software because malicious programs such as viruses, worms, trojans, spyware and ransomware can damage files, steal information, slow down the system, corrupt the operating system and spread to other computers. Antivirus software helps detect, block, quarantine and remove such malicious code.

Techniques used by antivirus programs:

- (i) **Signature-based detection:** It compares files with known virus signatures stored in the antivirus database.
- (ii) **Heuristic detection:** It checks suspicious behaviour or code patterns to detect new or modified malware.
- (iii) **Behaviour monitoring:** It watches program actions such as unauthorized file modification, registry changes or unusual network activity.
- (iv) **Sandboxing:** Suspicious programs are executed in an isolated environment to observe their behaviour safely.

1(g) Write short notes on E-learning and Security threats on Internet.

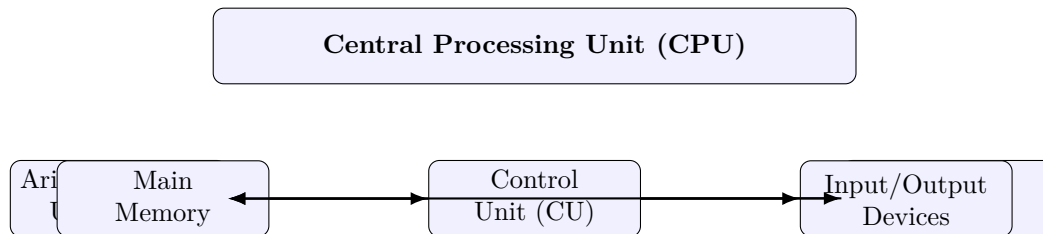
Answer. E-learning: E-learning means learning through electronic and internet-based tools. It includes online classes, recorded lectures, learning management systems, quizzes, discussion forums, assignments, digital libraries and video conferencing. Its benefits are flexibility, self-paced learning, easy distribution of study material, online assessment and wider access to education. Its limitations include dependence on internet, lack of face-to-face interaction and need for digital skills.

Security threats on Internet: Internet users face threats such as viruses, phishing, identity theft, hacking, spam, spyware, ransomware, fake websites, data leakage and denial-of-service attacks. These threats can steal passwords, damage data, misuse bank details and compromise privacy. Protection methods include antivirus software, firewall, strong passwords, two-factor authentication, software updates, secure websites using HTTPS and careful handling of email links and attachments.

Question 2

2(a) What is CPU? Describe the structure of the CPU and explain its components with the help of a diagram.

Answer. CPU stands for Central Processing Unit. It is the main processing unit of a computer and is often called the brain of the computer. It fetches instructions from memory, decodes them, executes them and controls the operation of all other parts of the computer.



Components of CPU:

- (i) **Arithmetic Logic Unit (ALU):** Performs arithmetic operations such as addition and subtraction, and logical operations such as comparison, AND, OR and NOT.
- (ii) **Control Unit (CU):** Directs and coordinates all operations. It fetches instructions, decodes them and sends control signals to other components.
- (iii) **Registers:** Small, high-speed storage locations inside the CPU. They temporarily hold instructions, addresses and data during processing.

2(b) What is Operating System? What kind of services does operating system provide to users and programs?

Answer. An operating system is system software that acts as an interface between the user, application programs and computer hardware. It manages resources and provides a convenient environment for program execution.

Services provided by operating system:

- (i) **User interface:** Provides command-line interface or graphical user interface.
- (ii) **Program execution:** Loads programs into memory and starts execution.
- (iii) **Process management:** Creates, schedules and terminates processes.
- (iv) **Memory management:** Allocates and deallocates memory to programs.
- (v) **File management:** Creates, deletes, stores and organizes files and folders.
- (vi) **Input/output management:** Controls devices through drivers and handles I/O operations.
- (vii) **Security and protection:** Provides login, permissions and access control.
- (viii) **Error detection:** Detects hardware and software errors and reports them.

2(c) How does data get transmitted via unguided transmission media? Explain radio, microwave and infrared modes of wireless transmission.

Answer. Unguided transmission media transmit data without physical wires. Signals travel through air or space as electromagnetic waves. A transmitter converts data into electromagnetic signals, and a receiver captures and converts them back into usable data.

Mode	Main characteristics	Uses
Radio waves	Omnidirectional waves, can travel long distances and may pass through walls. They are affected by interference.	Broadcasting, Wi-Fi, mobile communication.
Microwaves	High-frequency waves, usually line-of-sight and directional. They offer high bandwidth.	Satellite links, cellular towers, long-distance communication.
Infrared	Short-range waves that usually require line-of-sight and cannot pass through walls.	Remote controls, short-range device communication.

Question 3

3(a) What are Optical Memories? Explain briefly any four types of optical disks.

Answer. Optical memories are secondary storage media in which data is read or written using laser light. They store data as tiny pits and lands or marks on a reflective surface. Optical disks are portable, inexpensive and useful for distribution and backup.

Types of optical disks:

- (i) **CD-ROM:** Compact Disk Read Only Memory. Data is pre-recorded and can only be read. It is used for software and multimedia distribution.
- (ii) **CD-R:** Compact Disk Recordable. Data can be written once and read many times.
- (iii) **CD-RW:** Compact Disk Rewritable. Data can be erased and rewritten many times.
- (iv) **DVD:** Digital Versatile Disk. It has higher capacity than CD and is used for movies, software and large data storage.
- (v) **Blu-ray Disk:** Uses blue-violet laser and stores much more data than DVD. It is used for high-definition video and large backups.

3(b) What is cloud computing? Explain the various cloud services and how they benefit organizations. Describe key features and benefits of cloud-based solutions.

Answer. Cloud computing is the delivery of computing resources such as servers, storage, databases, software and networking over the internet on demand. Instead of owning all infrastructure, organizations can use services provided by cloud providers and pay according to usage.

Cloud service models:

- (i) **IaaS (Infrastructure as a Service):** Provides virtual machines, storage and networks. Example use: hosting applications without buying servers.
- (ii) **PaaS (Platform as a Service):** Provides a platform for developing, testing and deploying applications. It includes runtime, database and development tools.
- (iii) **SaaS (Software as a Service):** Provides ready-to-use software through a browser. Examples include email, online office tools and CRM systems.

Key features and benefits:

- On-demand self-service and quick provisioning.
- Scalability according to workload.
- Pay-as-you-use cost model.
- Access from anywhere through internet.

- Reduced need for local infrastructure and maintenance.
- Backup, disaster recovery and easier collaboration.
- Automatic updates and improved resource utilization.

3(c) What are open source software and why are they in demand? List the key criterion of OSD compliance.

Answer. Open source software is software whose source code is available to users for study, modification and redistribution under an open source licence. It is in demand because it reduces cost, encourages community development, improves transparency, allows customization and avoids dependence on a single vendor.

Key criteria of Open Source Definition (OSD) compliance include:

- (i) Free redistribution of the software.
- (ii) Source code must be available.
- (iii) Derived works and modifications must be allowed.
- (iv) No discrimination against persons or groups.
- (v) No discrimination against fields of use.
- (vi) Licence must not be specific to a product.
- (vii) Licence must not restrict other software.
- (viii) Rights must apply to all recipients.

Question 4

4(a) Discuss the key characteristics of printers on the basis of quality of print.

Answer. Printer quality is judged by the clarity, sharpness and accuracy of printed output. Important characteristics are:

- (i) **Resolution:** Measured in dpi (dots per inch). Higher dpi gives sharper output.
- (ii) **Print technology:** Laser printers usually give sharp text; inkjet printers give good colour and photo output; dot matrix printers give lower quality.
- (iii) **Colour quality:** Ability to reproduce accurate colours, gradients and photographs.
- (iv) **Font and graphics quality:** Smooth curves, clear edges and proper spacing improve readability.
- (v) **Paper handling:** Better paper support improves final output quality.
- (vi) **Speed versus quality:** Draft mode is faster but lower quality; high quality mode is slower but clearer.

4(b) What are utility softwares? Explain Disk Defragmenter, CHKDSK, Data compression and Disk management.

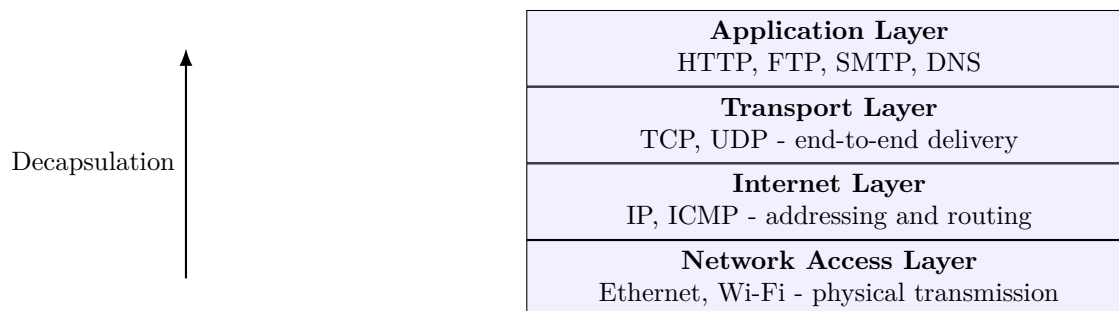
Answer. Utility software is system software designed to maintain, analyze, optimize, protect and manage a computer system. It supports the operating system and improves reliability and performance.

- (i) **Disk Defragmenter:** Rearranges fragmented files on a magnetic hard disk so that file parts are stored in contiguous locations. This improves access speed on HDDs. It is generally not required for SSDs.

- (ii) **CHKDSK:** A disk checking utility that scans file system errors, bad sectors and logical disk problems. It can repair file system inconsistencies.
- (iii) **Data compression:** Reduces the size of files or folders to save storage space and make transfer faster. Examples include ZIP and RAR compression.
- (iv) **Disk management:** Helps create, delete, format and resize disk partitions. It also assigns drive letters and manages volumes.

4(c) Describe the TCP/IP model with the help of a diagram.

Answer. TCP/IP is the standard protocol suite used for internet communication. It defines how data is divided into packets, addressed, transmitted, routed and received across networks.



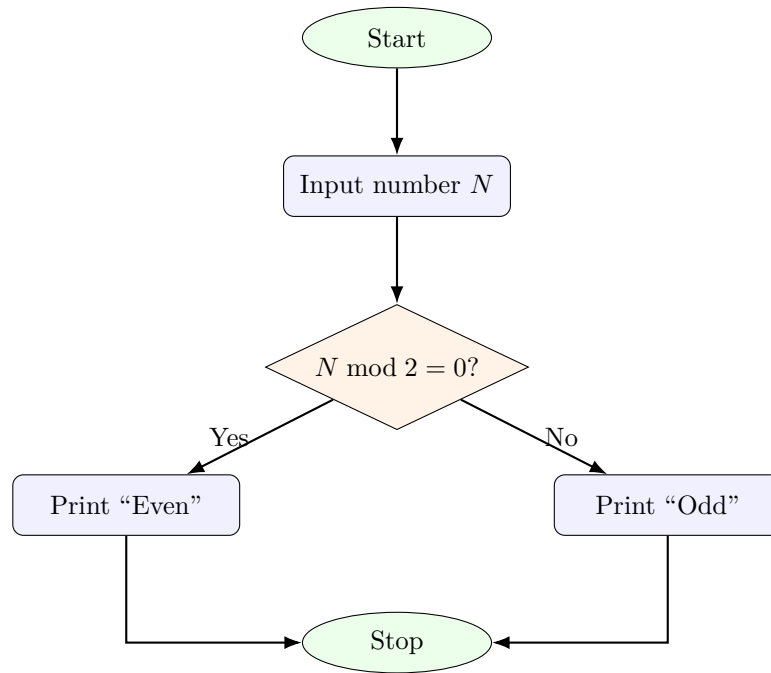
Layers:

- (i) **Application layer:** Provides services to user applications such as web, email and file transfer.
- (ii) **Transport layer:** Provides process-to-process communication. TCP gives reliable connection-oriented service; UDP gives faster connectionless service.
- (iii) **Internet layer:** Handles logical addressing and routing using IP addresses.
- (iv) **Network access layer:** Handles actual transmission over physical media such as Ethernet or Wi-Fi.

Question 5

5(a) Draw a flowchart to find whether a given number is even or odd.

Answer. A number is even if it is exactly divisible by 2. In programming form, if $N \bmod 2 = 0$, then N is even; otherwise it is odd.



5(b) What is software project management? Explain various project management activities.

Answer. Software project management is the process of planning, organizing, monitoring and controlling software development so that the project is completed within required time, cost and quality constraints. It coordinates people, tasks, tools and resources.

Major project management activities:

- (i) **Project planning:** Define objectives, scope, deliverables and milestones.
- (ii) **Requirement management:** Collect, analyze and control user requirements.
- (iii) **Estimation:** Estimate cost, time, effort and required resources.
- (iv) **Scheduling:** Break the project into tasks and assign deadlines.
- (v) **Resource allocation:** Assign people, hardware, software and budget.
- (vi) **Risk management:** Identify possible risks and prepare mitigation plans.
- (vii) **Monitoring and control:** Track progress and compare it with the plan.
- (viii) **Quality management:** Ensure testing, review and quality standards.
- (ix) **Communication management:** Maintain communication among team, users and stakeholders.
- (x) **Documentation and reporting:** Prepare project documents, status reports and final delivery records.

5(c) What is mail merge? Explain various steps of mail merge with the help of an example.

Answer. Mail merge is a word processing feature used to create many personalized documents from one common template and a data source. It is useful for letters, certificates, invitations, labels and envelopes.

Example: A college wants to send fee reminder letters to 500 students. The letter body is the same, but student name, roll number, fee amount and address are different. Mail merge automatically inserts each student's data into the letter template.

Steps of mail merge:

- (i) **Create the main document:** Write the common letter or certificate.
- (ii) **Create/select data source:** Prepare a table or spreadsheet containing fields such as Name, Roll No., Address and Amount.
- (iii) **Insert merge fields:** Place fields like «Name», «RollNo» and «Amount» in the document.
- (iv) **Preview results:** Check whether the correct data appears in the right place.
- (v) **Finish and merge:** Generate separate personalized letters, print them or send them by email.

Final exam note: In this paper, Question 1 is compulsory and any three questions from Questions 2-5 are to be attempted. This booklet gives solutions for all questions for full preparation.